Propagator Eval — Simple Experiment Explainer

What each experiment is for, and what result we are trying to learn

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Big picture. This project studies how different propagation operators recover missing cross-client node features in communication-free subgraph federated learning. The goal is not just to report accuracy, but to understand *which operators work, why they work, when they scale, and when they fail.*

Main research questions

- Which propagation operator best reconstructs missing features?
- How quickly and stably does each operator converge?
- Do better intrinsic recovery metrics translate into better downstream GNN accuracy?
- Which operators remain practical at larger scale?
- What happens when the graph is heterophilic and homophily assumptions weaken?

Operators we compare

Operator	What it is	Why it matters
Adjacency	Classical FedProp-style normalized adjacency propagation	Baseline harmonic extension method
Diffusion	Taylor / diffusion-based propagation	Strong existing communication-free reference
Chebyshev	Polynomial heat-kernel approximation	Tests whether a richer smoother improves quality or speed
APPNP	Personalized PageRank-style propagation	Tests a non-Dirichlet fixed point with teleportation
Random Walk	Asymmetric D^-1 A propagation	Tests another non-Dirichlet operator baseline
Heat Kernel exact	Exact reference on small graphs only	Gives a theoretical quality reference for small datasets

Datasets and why they exist

- **Cora:** anchor dataset for deep understanding
- **Citeseer, Pubmed:** homophilic reproduction datasets
- **OGBN-Arxiv:** scalability dataset
- **Texas, Wisconsin:** heterophily stress-test datasets

Experiment phases

Phase 1 — Cora intrinsic core

Purpose. Compare all propagation operators on Cora only, before scaling to other datasets.

What we measure.

- MSE
- cosine similarity
- recovery ratio
- boundary coverage
- iteration count
- residual curves
- convergence flag
- wall-clock time

What we are trying to find out.

- Which operator reconstructs missing features best on a clean homophilic benchmark?
- Which operators are stable and efficient?
- Do the operators separate clearly enough to build the main paper story?

What a useful result looks like. A ranking of operators on Cora, plus clear plots showing quality, convergence, and runtime differences.

Phase 2 — Cora ablations

Purpose. Test whether the Cora conclusions are robust or just artifacts of one setting.

What we vary.

- APPNP alpha
- convergence tolerance
- hop depth
- other Cora-specific sensitivity settings

What we are trying to find out.

- Which knobs actually matter?
- Which defaults should be frozen before we reproduce on other datasets?
- Are some operators fragile while others are robust?

What a useful result looks like. A small set of recommended default settings and a short explanation of which hyperparameters materially change conclusions.

Phase 3 — Cora downstream evaluation

Purpose. Connect intrinsic propagation quality to downstream GCN/GAT performance on the same anchor dataset.

What we measure.

- downstream accuracy
- gap closed relative to zero-hop and oracle
- joined intrinsic/downstream tables
- runtime/accuracy tradeoffs

What we are trying to find out.

- Does better feature recovery actually help learning?
- Is intrinsic quality predictive of downstream benefit?
- Are some operators intrinsically strong but downstream weak, or vice versa?

What a useful result looks like. A clear link between intrinsic recovery metrics and downstream performance, ideally with a scatter plot and a gap-closed summary table.

Phase 4 — Homophilic reproduction

Purpose. Reproduce the Cora story on Citeseer and Pubmed using the frozen protocol.

What we are trying to find out.

- Does the best Cora operator remain strong on similar citation graphs?
- Are operator rankings stable across homophilic datasets?
- Are any failures dataset-specific?

What a useful result looks like. Cross-dataset tables showing whether the Cora findings generalize to other homophilic settings.

Phase 5 — Scalability on OGBN-Arxiv

Purpose. Check which operators remain practical at larger scale.

What we emphasize.

- runtime
- convergence
- memory/resource behavior
- quality per unit time

What we are trying to find out.

- Which operators are still usable on a much larger graph?
- Which methods become too slow or too memory-heavy?
- Are failures themselves scientifically informative?

What a useful result looks like. A runtime-quality tradeoff story, including documented failures if some operators do not scale well.

Phase 6 — Heterophily stress test

Purpose. Test what happens when the homophily assumption weakens.

What we are trying to find out.

- Does propagation help, hurt, or become unstable on heterophilic graphs?
- Do non-Dirichlet operators behave differently from Dirichlet ones?
- Can low intrinsic recovery predict low downstream gains or outright harm?

What a useful result looks like. A clear statement about the limits of propagation under heterophily, not just its strengths on citation graphs.

How the paper story should emerge

The final paper should read like this:

Cora explains the phenomenon → Citeseer/Pubmed show it reproduces → OGBN-Arxiv shows what scales → Texas/Wisconsin show where propagation breaks or weakens

Bottom line

Each phase answers a different scientific question:

- **Phase 1:** what works on the anchor dataset
- **Phase 2:** what is robust
- **Phase 3:** what helps downstream learning
- **Phase 4:** what generalizes
- **Phase 5:** what scales
- **Phase 6:** what fails under heterophily

Together, they give a full picture of operator quality, mechanism, usefulness, and limits.